Predicting Cardiovascular Risk using Electronic Health Records

UE24CS352A Machine Learning - Mini-Project | Team 25 | Madhumitha (PES1UG24AM923), Anjali Arun (PES1UG24AM421)

1. Problem statement

Cardiovascular disease (CVD) is the leading cause of death worldwide, and identifying at-risk patients early enables prevention. We predict cardiovascular risk from routinely recorded clinical data (binary classification) and ask two questions: which models work best, and do the conclusions hold across two quite different datasets? Following Nguyen (Stanford CS229) on EHR-based CVD risk prediction, we compare interpretable regularised logistic regression with tree ensembles. The original STRIDE EHR data is private, so the project is split into **Study A** (prevalent CVD, large balanced data) and **Study B** (10-year incident CHD, small imbalanced data with missing values), which together cover the paper's setting.

2. Datasets

	Study A: Kaggle Cardiovascular Disease	Study B: Framingham Heart Study
Task / label	CVD present now (prevalent)	CHD within 10 years (incident)
Records	70,000 raw -> 67,941 after cleaning	4,238 (none dropped)
Features	13 (adds BMI, pulse pressure)	15 (smoking, diabetes, BP meds, cholesterol, ...)
Positive class	49.7% (balanced)	15.2% (imbalanced)
Data issues	2,059 implausible rows (BP, height, weight) removed; no missing values	582 rows (14%) with missing values; median-imputed in-pipeline
Train / test	54,352 / 13,589	3,390 / 848

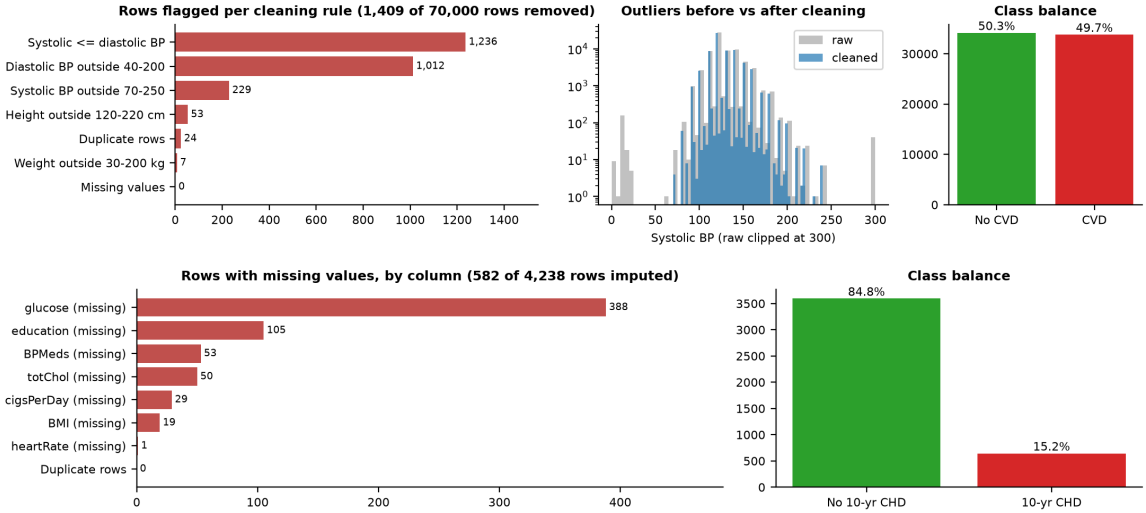


Fig. 1. Data-quality reports. Top: Study A (rows flagged per rule, outliers before/after cleaning, class balance). Bottom: Study B (missing values per column, class balance).

3. Approach and protocol (identical for both studies)

Models: logistic regression with Ridge, Lasso and Elastic Net penalties (standardised features); Random Forest; and **XGBoost** as the main boosted model. **Protocol:** stratified 80/20 split (seed 42); median imputation and scaling fitted inside each cross-validation fold (no leakage); 10-fold grid search on AUROC using the training data only; one evaluation on the held-out test set. Study B uses class weighting for the 15% positive class. **Metrics:** AUROC (primary), average precision, accuracy, sensitivity, specificity at threshold 0.5, confusion matrices, calibration, and AUROC within gender and age subgroups.

4. Implementation

Python (pandas, scikit-learn, XGBoost, matplotlib). `src/benchmark/` holds the shared pipeline: `datasets.py` (loading, cleaning, per-dataset settings), `train.py`, `visualize.py`, `make_report.py`, `make_slides.py`. The Framingham exploration and first baselines (data inspection, quality report, untuned LR/RF, coefficients) are the scripts in `src/`; the benchmark reuses their exact split. Commands are in the README.

5. Results

	Study A (Kaggle, prevalent CVD)					Study B (Framingham, 10-yr CHD)				
Model	CV AUC	Test AUC	Avg prec	Sens	Spec	CV AUC	Test AUC	Avg prec	Sens	Spec
LogReg (Ridge)	0.793	0.783	0.765	0.67	0.77	0.733	0.700	0.300	0.58	0.68
LogReg (Lasso)	0.793	0.784	0.766	0.67	0.77	0.733	0.699	0.290	0.59	0.68
LogReg (ElasticNet)	0.793	0.784	0.766	0.67	0.77	0.734	0.699	0.290	0.60	0.68
Random Forest	0.801	0.793	0.779	0.68	0.77	0.718	0.684	0.272	0.54	0.69
XGBoost (GBM)	0.802	0.796	0.783	0.69	0.76	0.707	0.676	0.278	0.52	0.69

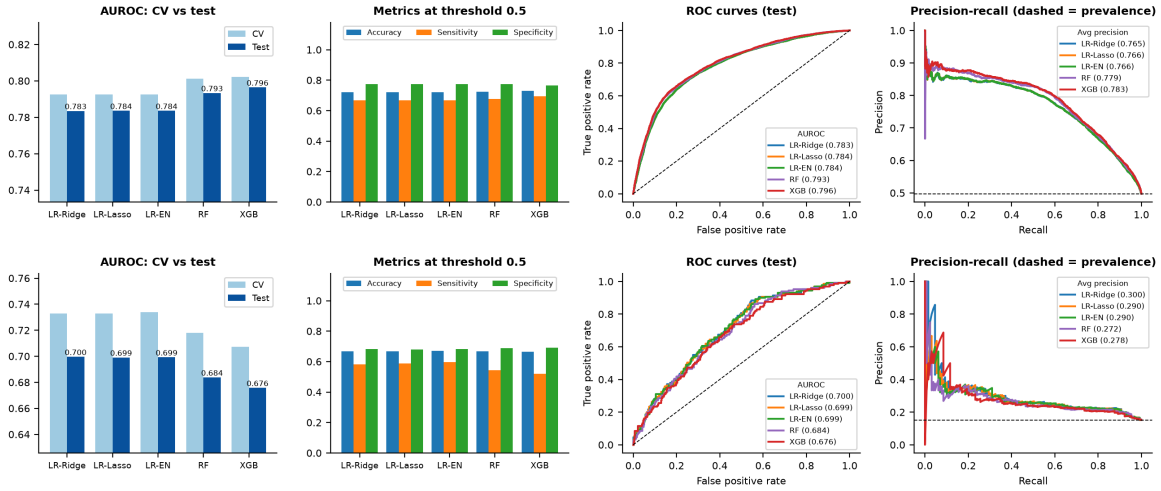


Fig. 2. Model comparison (top: Study A, bottom: Study B): CV vs test AUROC, metrics at threshold 0.5, ROC and precision-recall curves.

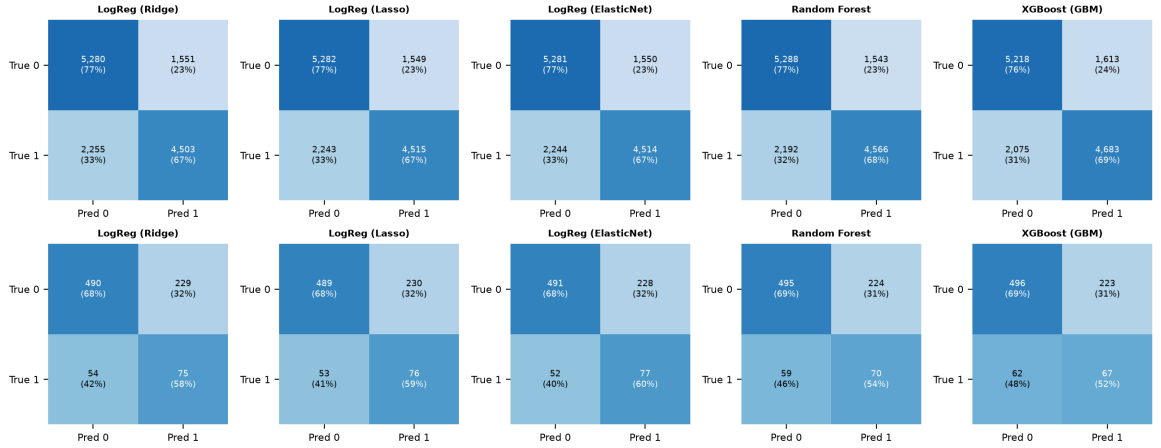


Fig. 3. Test-set confusion matrices (threshold 0.5, row-normalised percentages). Class weighting in Study B trades specificity for sensitivity.

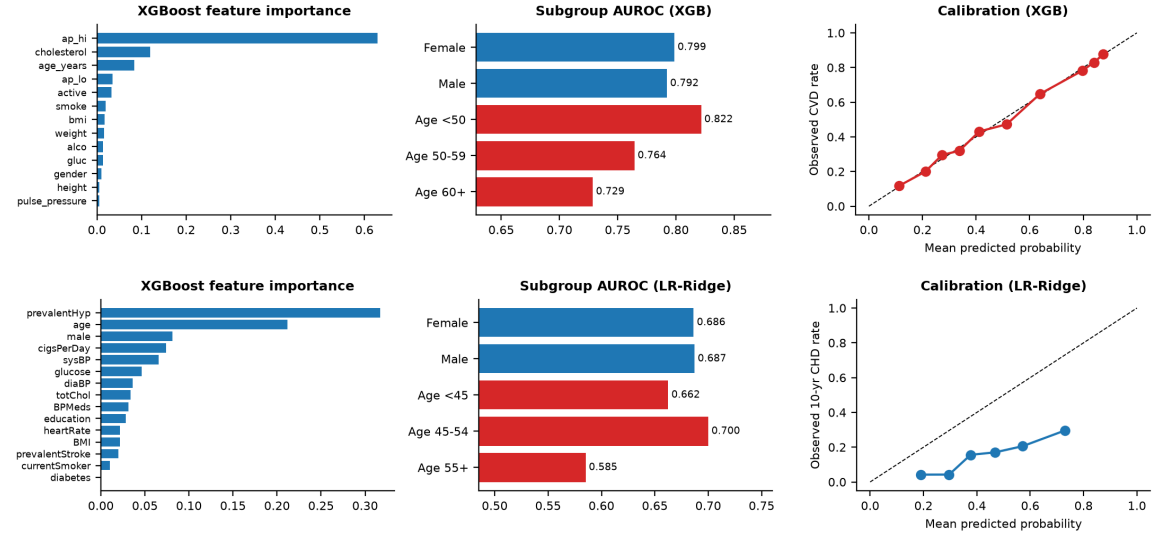


Fig. 4. Feature importance of the main tree model, subgroup AUROC and calibration of the best model (top: Study A, bottom: Study B).

6. Discussion and conclusions

Study A: best model XGBoost (GBM) (test AUROC 0.796); best logistic regression 0.784; XGBoost 0.796. **Study B:** best model LogReg (Ridge) (test AUROC 0.700); best logistic regression 0.700; XGBoost 0.676. CV and test AUROC differ by 0.006 (A) and 0.033 (B) for the best models; with only 848 test patients (129 events), Study B differences of a few hundredths are within noise. Absolute AUROC is lower in Study B (0.700 vs 0.796), plausibly because it predicts future events from a small cohort with fewer, noisier signals. Our first untuned Study B baselines (class-weighted LR / RF) reached AUROC 0.699 / 0.668 on the same test split, consistent with the tuned results above.

Fairness and calibration. Subgroup AUROC (A: gender 0.792-0.799; age 0.729-0.822 (weakest: Age 60+, 0.729); B: gender 0.686-0.687; age 0.585-0.700 (weakest: Age 55+, 0.585)) shows that performance is not uniform and should be audited before clinical use. In Study B, class weighting raises predicted probabilities (mean 0.44 vs observed rate 0.15), so rankings (AUROC) are valid but probabilities need recalibration before being read as risks.

Limitations / future work: Study A labels are prevalent disease with noisy self-reported fields and no ethnicity, so the paper's race-based fairness analysis could not be repeated; Study B is small, from one cohort, with a single random split. Next: repeated CV / confidence intervals, probability calibration, survival-style modelling of incident risk and fairness-aware training.